

Socket Programming Project 2 – TCP/IP Programming

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1. Provide real-time status updates of all connected monitoring services, including their operational status (i.e. validating task information, idle, etc.) and any alerts (i.e. “Online”, “Offline”, etc.) or anomalies detected (i.e. “Reconnecting”, etc.).	16
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1. Monitoring Service must use unique identifier (R2.1) to perform result reporting with Management Service. 34
 2. Send the results of service tests back to the management service immediately for real-time reporting. 34
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 1. Monitoring Service must be able to allow for the Management Service to reconnect without needing to restart the Monitoring Service. 37

How to Run the Programs

All Run Options

Monitoring_Service.py	Management_Service.py	UDP_echo_server.py
Terminal 1	Terminal 2	Terminal 3
(default Berlin Station) python Monitoring_Service.py	python Management_Service.py In prompt window type “Berlin” or “Hong Kong” (without quotes.)	Python UDP_echo_server.py
(Hong Kong Station) python Monitoring_Service.py “Hong Kong” “127.0.0.1” 10202 (With quotes. No quotes on port)		
(Berlin Station) python Monitoring_Service.py “Berlin” “127.0.0.1” 10101 (With quotes. No quotes on port)		
Visual Studio 2022	Visual Studio 2022	Visual Studio 2022
(default Berlin Station) Right click mouse button. Run without debug.	Right click mouse button. Run without debug. In prompt window type “Berlin” or “Hong Kong” (without quotes.)	Right click mouse button. Run without debug.

Fastest Setup Using Default Settings

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Use Visual Studio 2022

Right click mouse button.
Run without debug.

--

Use 3 terminals.

Terminal 1
(default Berlin Station)
python Monitoring_Service.py

Terminal 2

python Management_Service.py

In prompt window type "Berlin" (without quotes.)

Terminal 3

Python UDP_echo_server.py

Management Service Software Requirements Specification (SRS) (Abbreviated):

1. Management Service monitoring management

1. The Management Service must perform the existing tasks related to configuring and managing monitoring tasks.

Figure 1

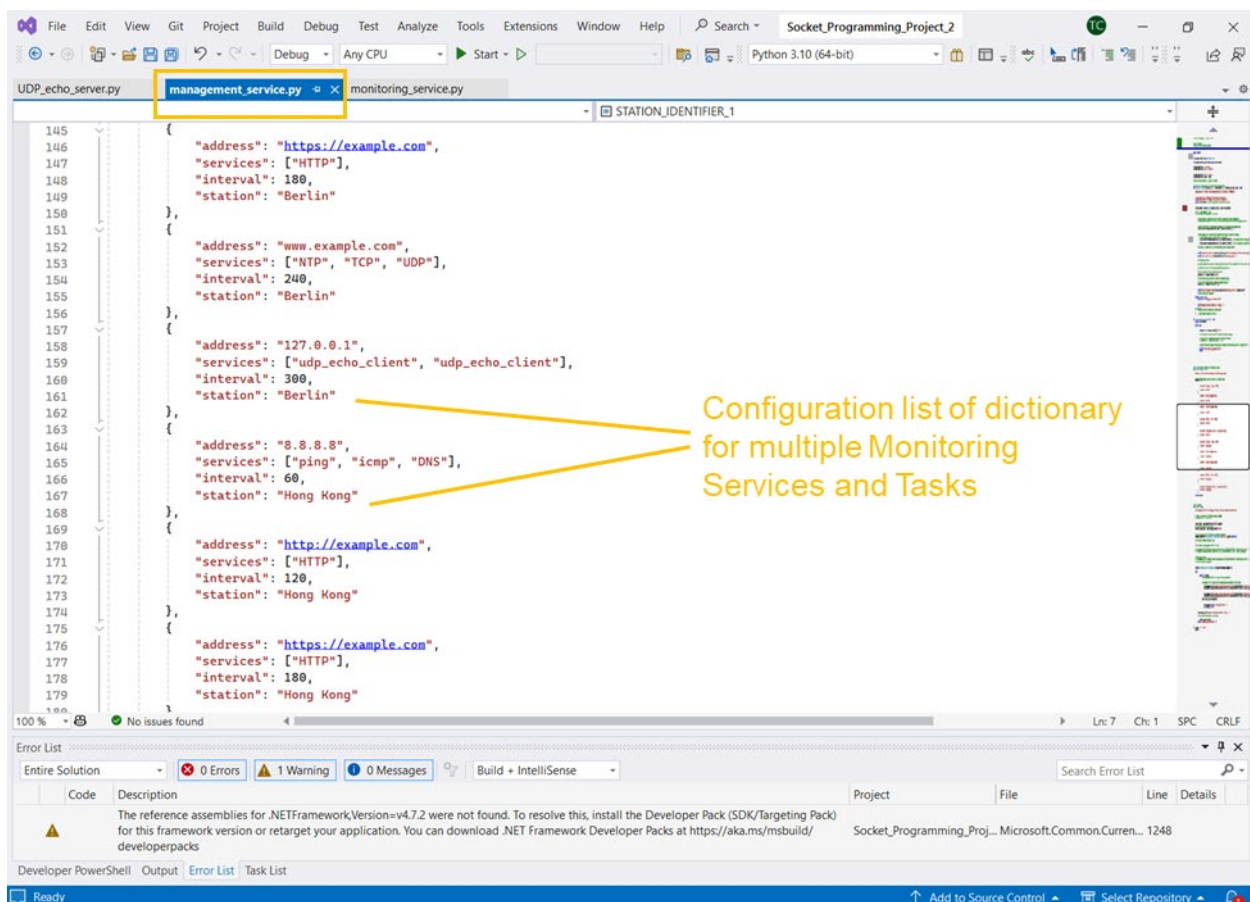
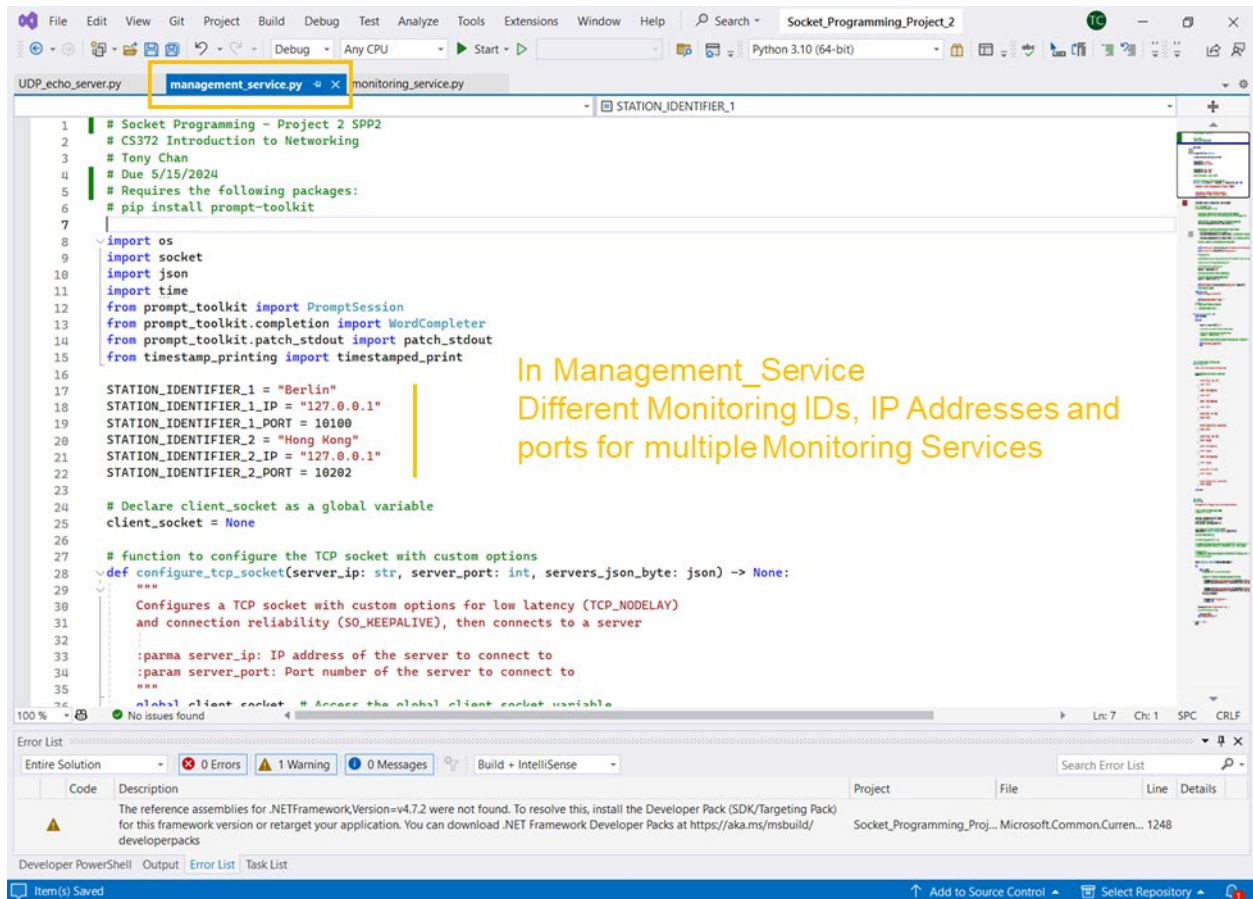


Figure 2



2. Connection Management

1. The management service must be capable of establishing and maintaining persistent TCP connections to multiple monitoring services, each identified by IP and port numbers.

Figure 3

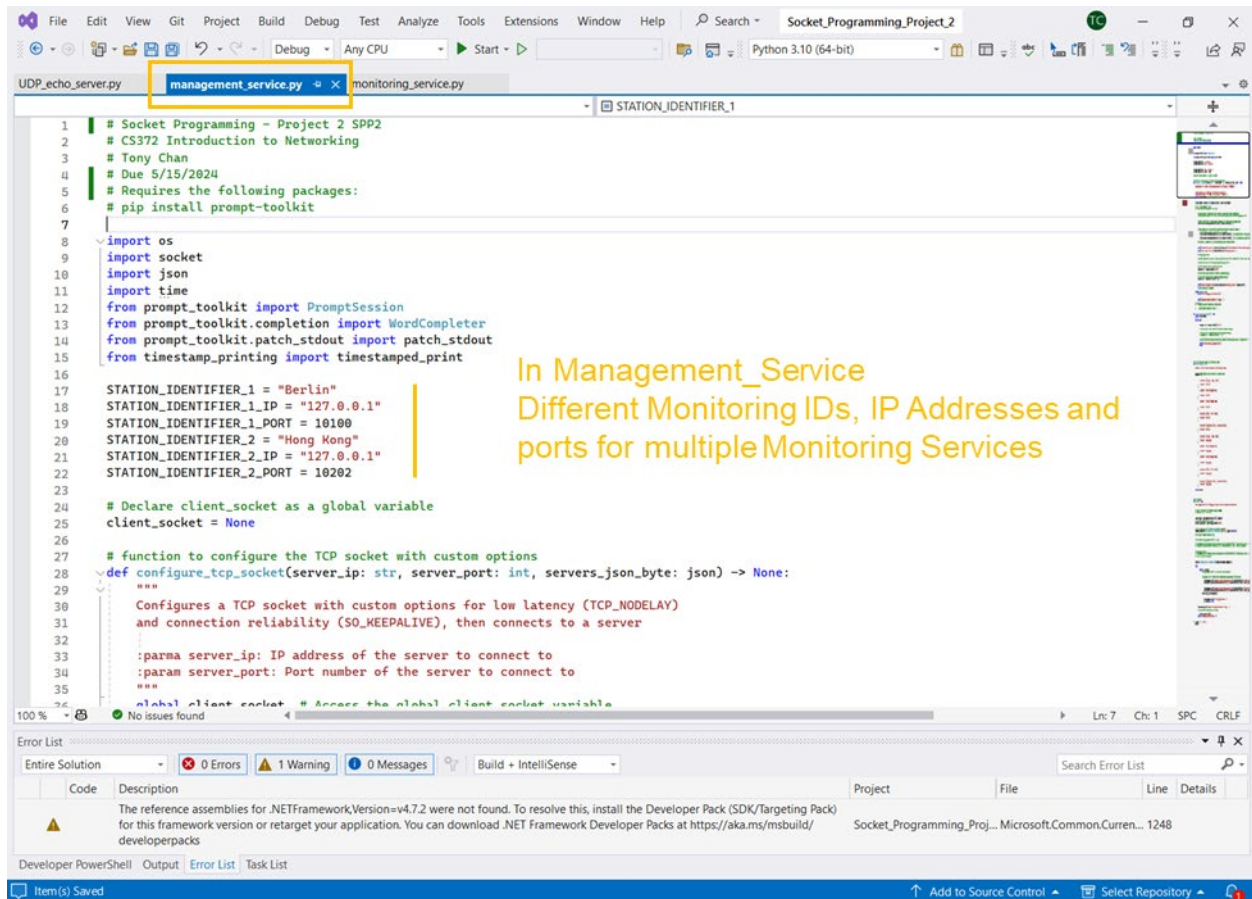
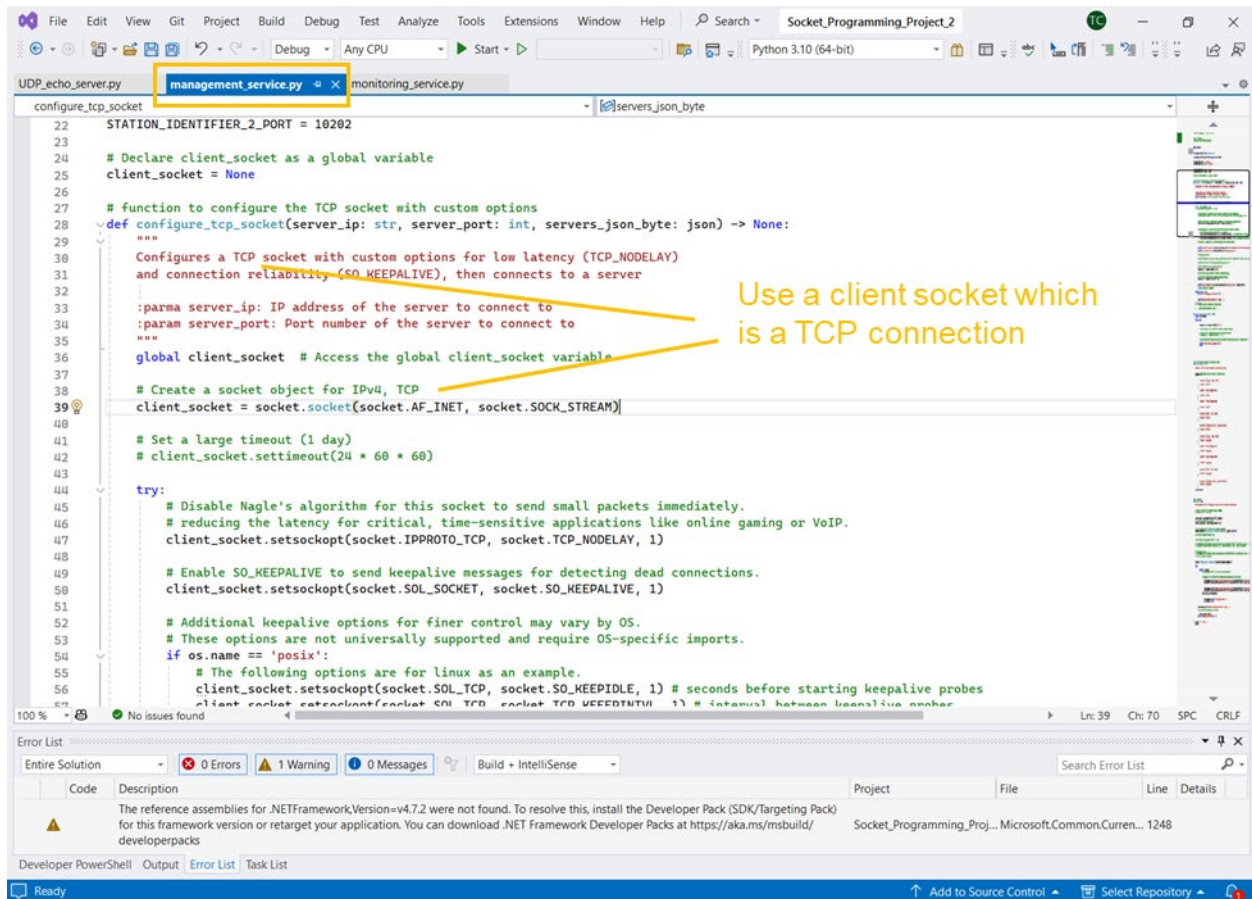


Figure 4



2. Implement TCP keepalive mechanisms to ensure long-term connection stability.

Figure 5

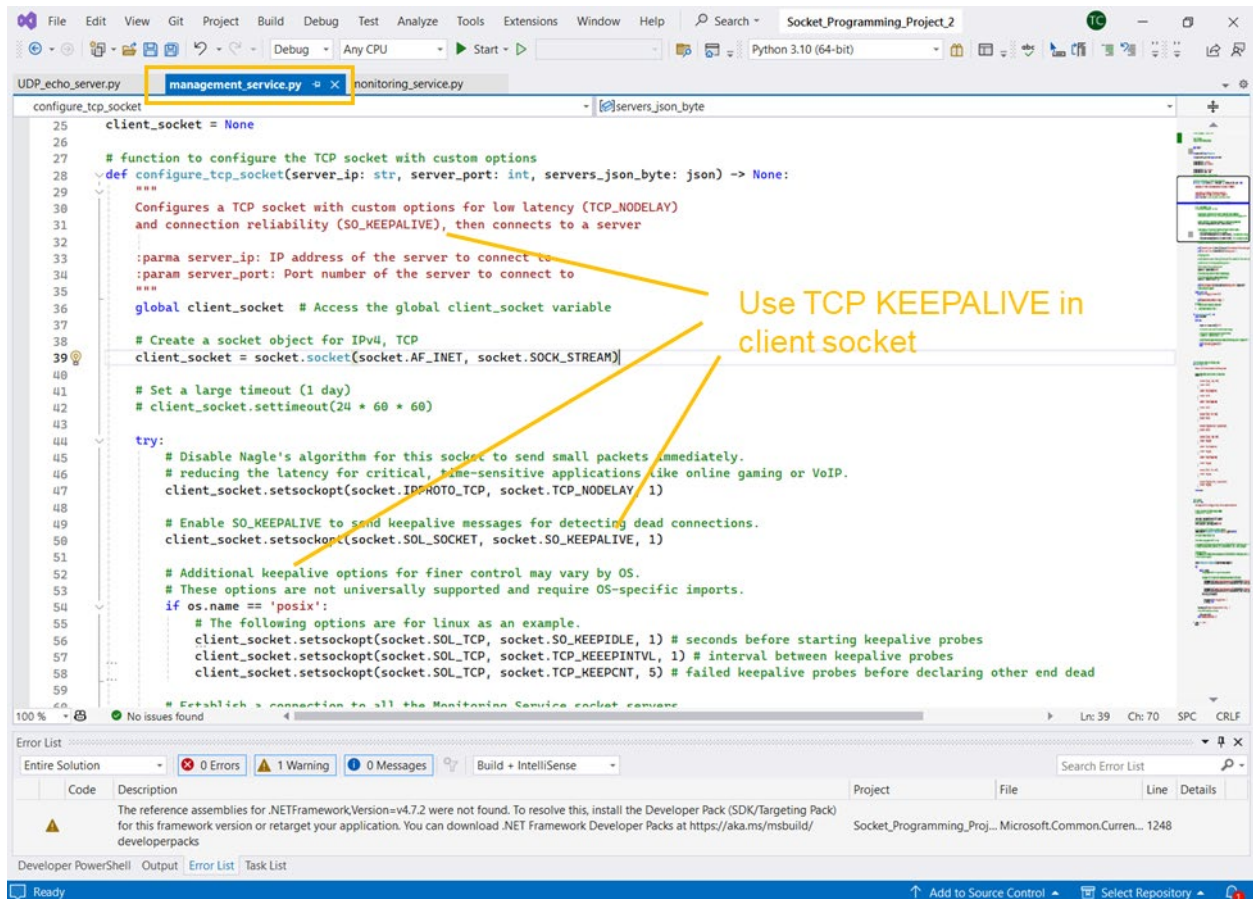
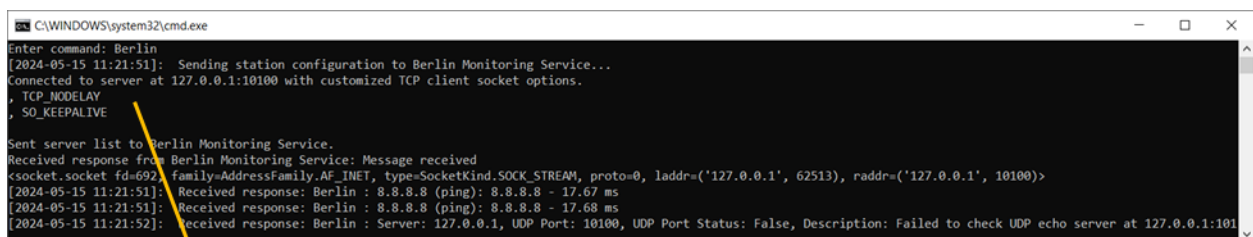


Figure 6



3. Automatically attempt reconnection if any connection to a monitoring service fails.

Figure 7

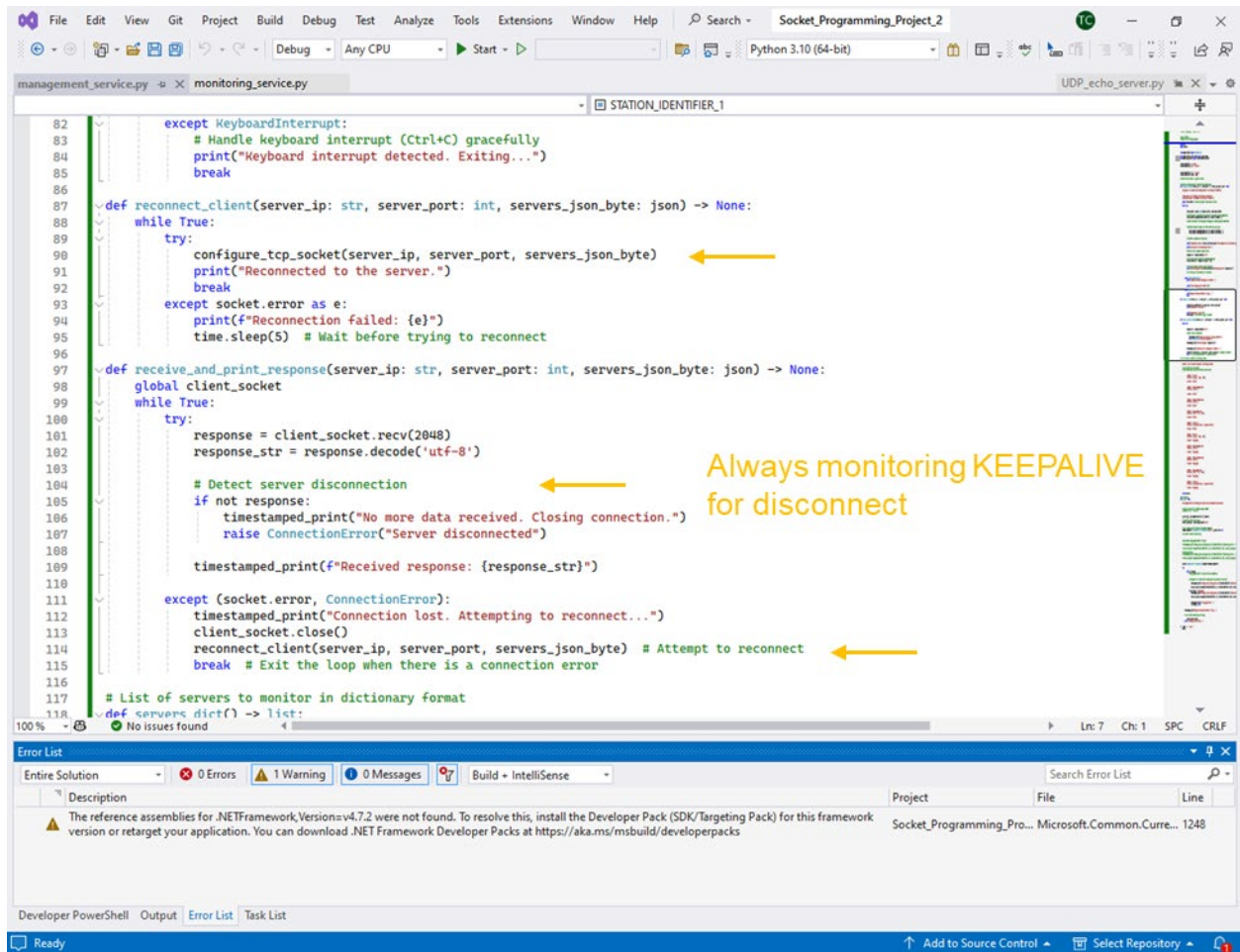


Figure 8

Management Service Monitoring Service is disconnected

```
C:\WINDOWS\system32\cmd.exe
[2024-05-16 13:48:33]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 17.14 ms
[2024-05-16 13:48:33]: Received response: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-16 13:48:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-16 13:48:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-16 13:48:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-16 13:48:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-16 13:48:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25', '74.6.231.20', '74.6.143.26']
[2024-05-16 13:49:33]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 17.42 ms
[2024-05-16 13:49:33]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.49 ms
[2024-05-16 13:49:34]: Received response: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['172.217.12.110']
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:814::200e']
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '74.6.143.26', '74.6.143.25', '98.137.11.164', '98.137.11.163', '74.6.231.21']
[2024-05-16 13:52:34]: No more data received. Closing connection.
[2024-05-16 13:52:34]: Connection lost. Attempting to reconnect...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
```

```
C:\WINDOWS\system32\cmd.exe
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:814::200e']
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-16 13:49:34]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '74.6.143.26', '74.6.143.25', '98.137.11.164', '98.137.11.163', '74.6.231.21']
[2024-05-16 13:52:34]: No more data received. Closing connection.
[2024-05-16 13:52:34]: Connection lost. Attempting to reconnect...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connection refused. Retrying in 5 seconds...
Connected to server at 127.0.0.1:10100 with customized TCP client socket options.
TCP_NODELAY
SO_KEEPALIVE
Sent server list to Monitoring Service.
Received response from Berlin Monitoring Service: Message received
Reconnected to the server.
Enter command:
```

It will continuously try to connect even when it gets disconnected. Once it gets connected the Management Service will send the Monitor configuration settings. Reconnection is an error handling issue.

4. Management Service must have a saved configuration that contains the list of Monitoring Services (e.g. can be a file, hard coded, or other persistent option).

Figure 9

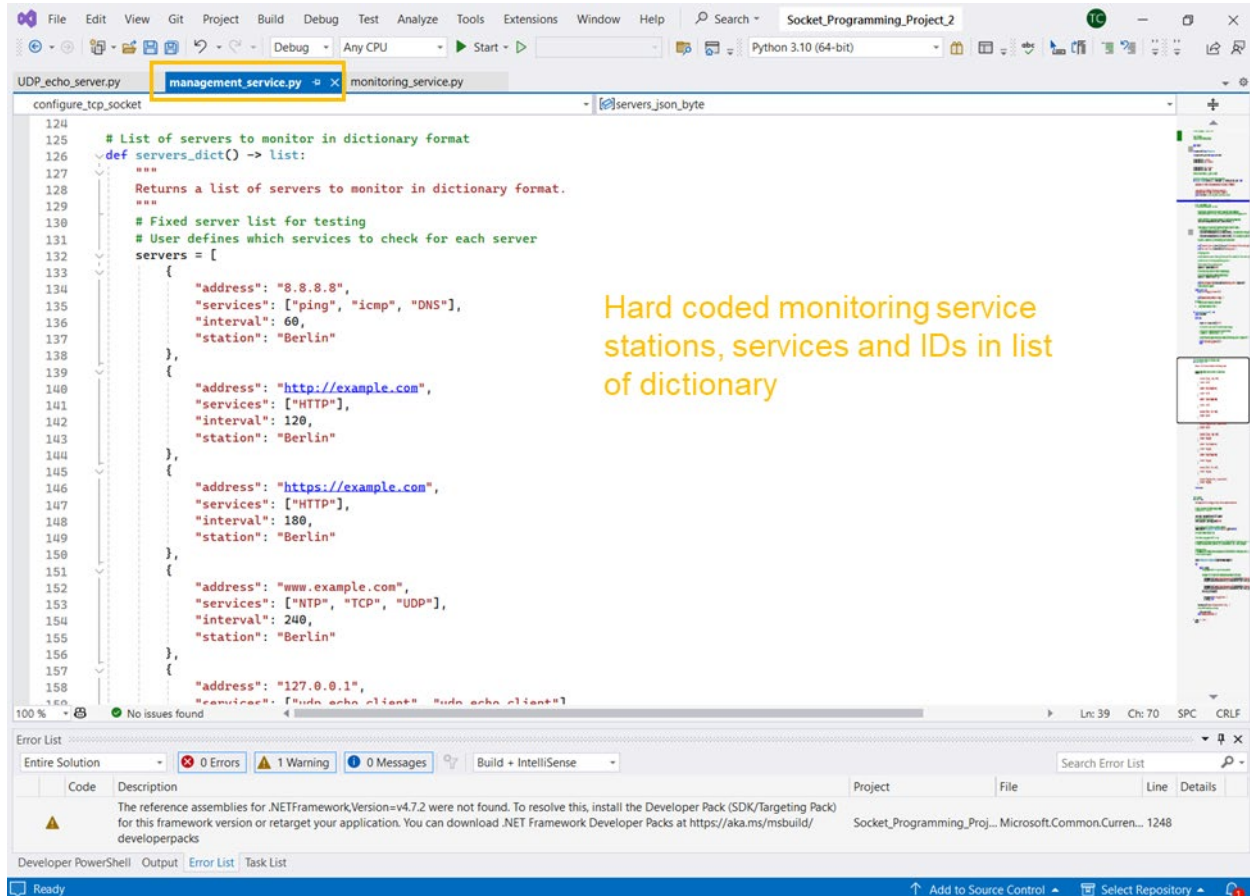
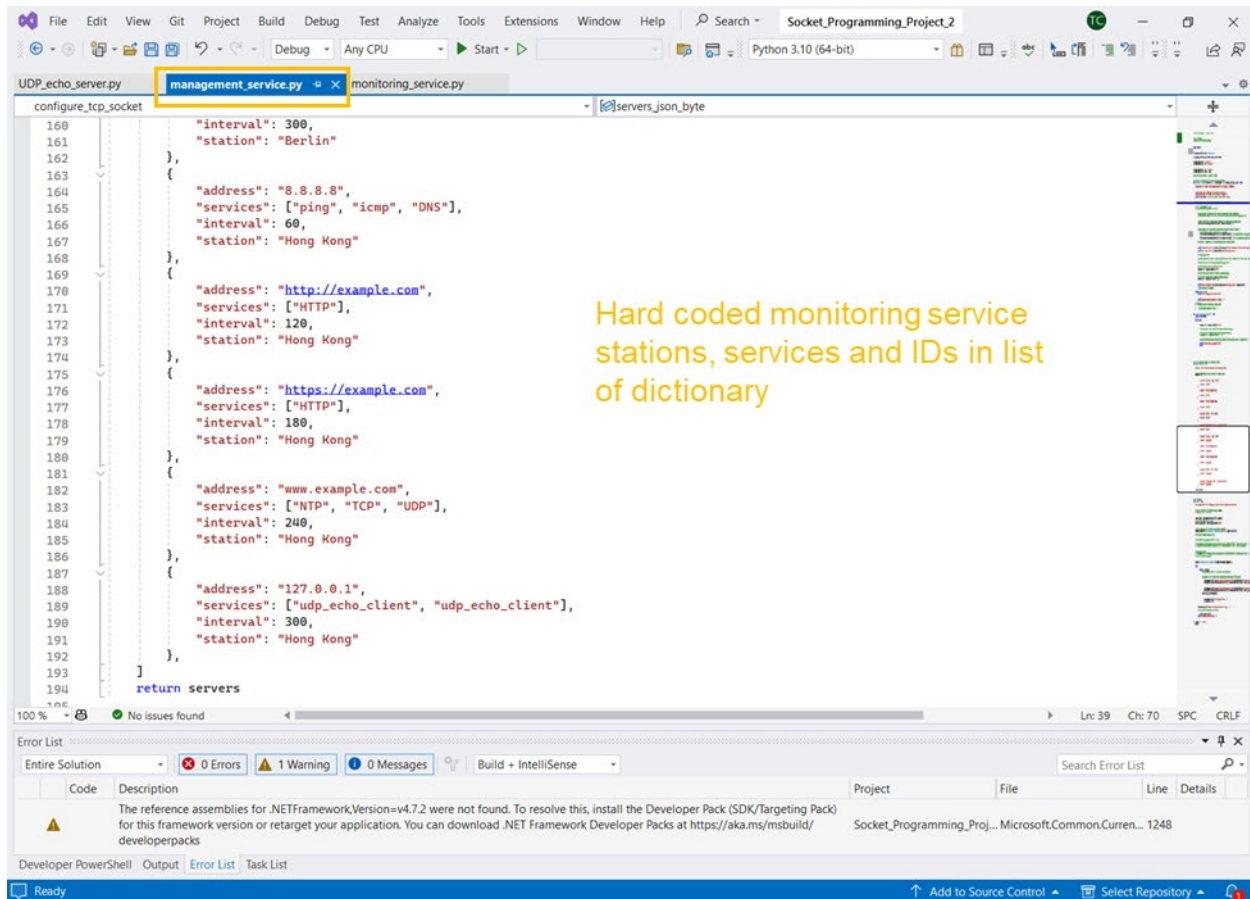


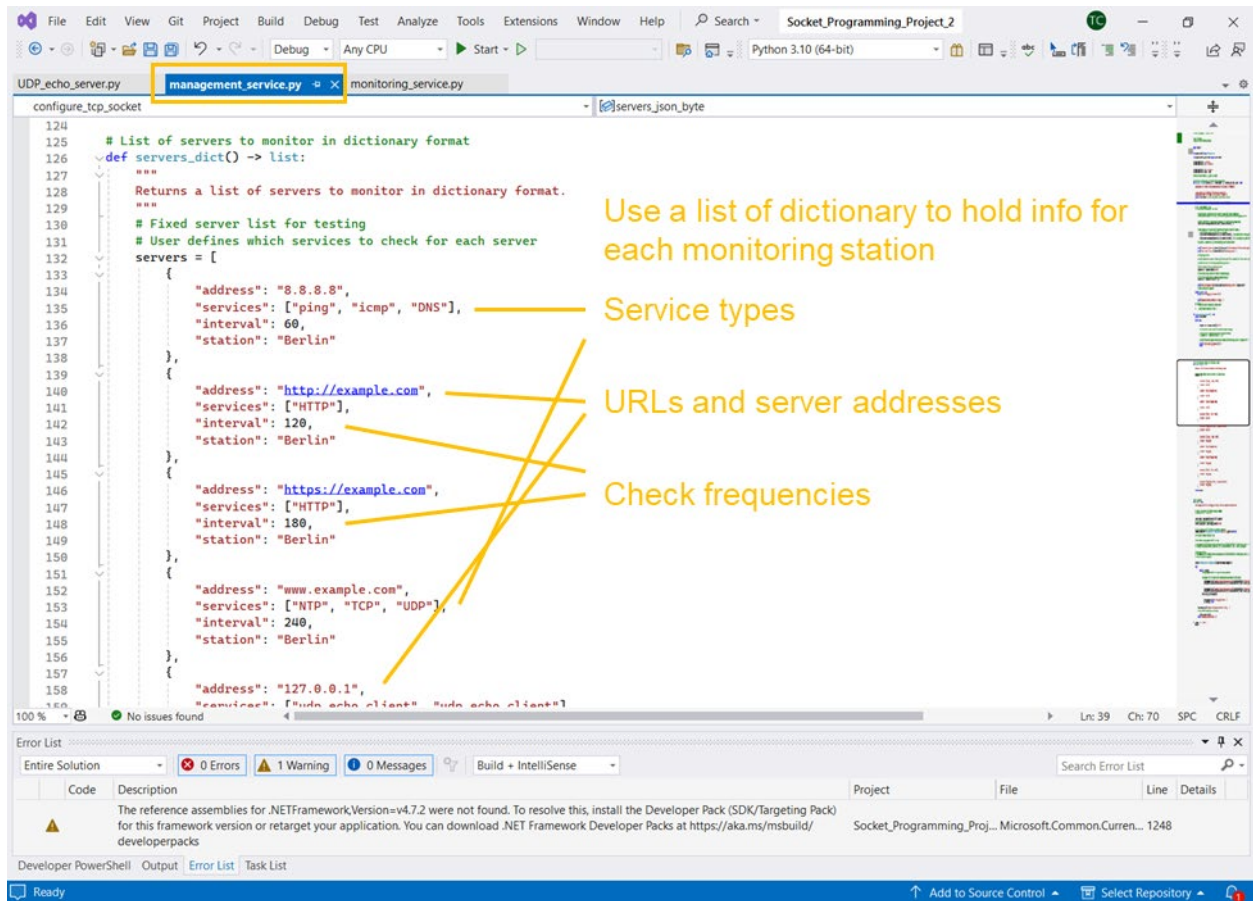
Figure 10



3. Configuration Management

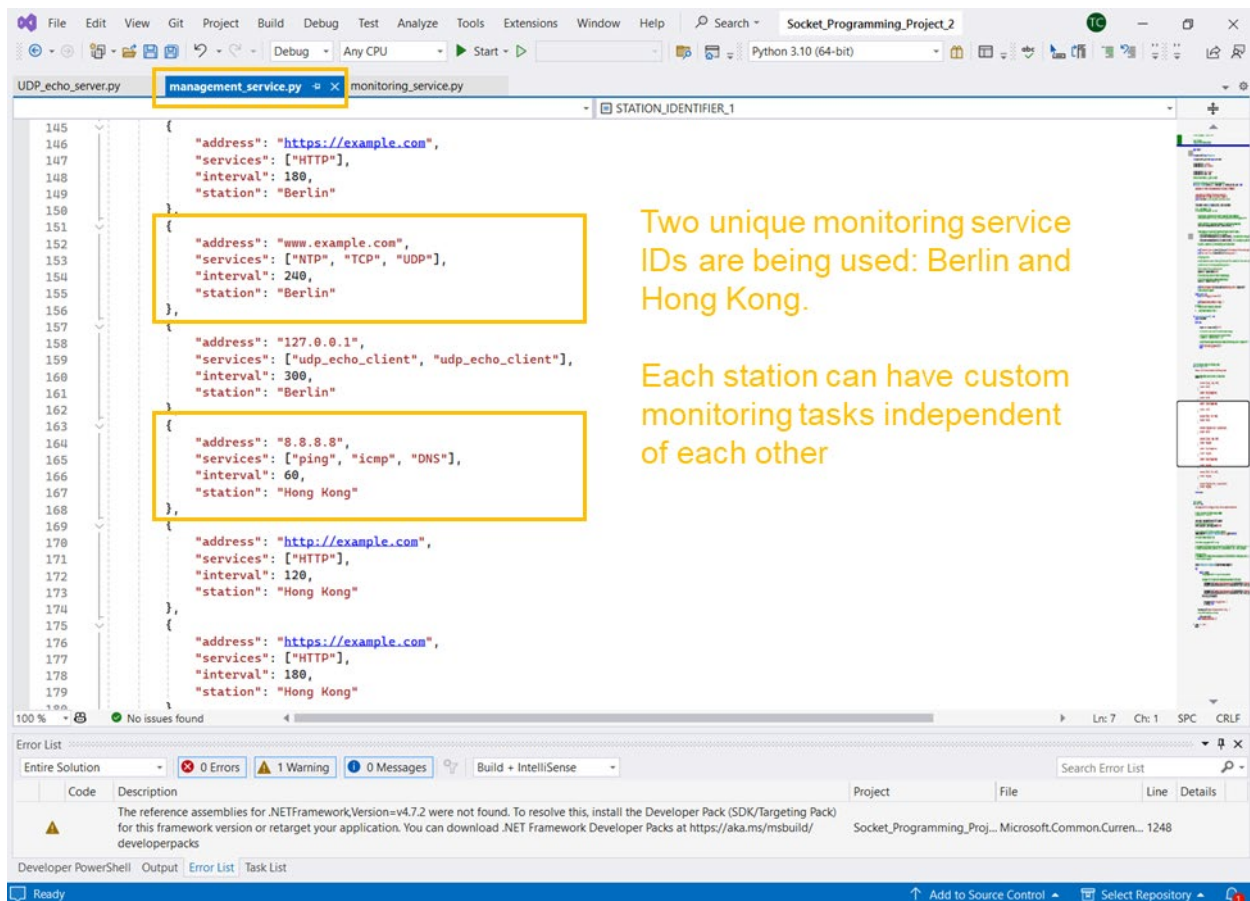
1. Hold all configuration information for monitoring tasks, including service types to monitor (HTTP, HTTPS, ICMP, DNS, NTP, TCP, UDP), parameters for checks (URLs, server addresses, port numbers), and check frequencies.

Figure 11



2. Assign and manage unique IDs for each monitoring service, linking them to specific monitoring tasks.

Figure 12



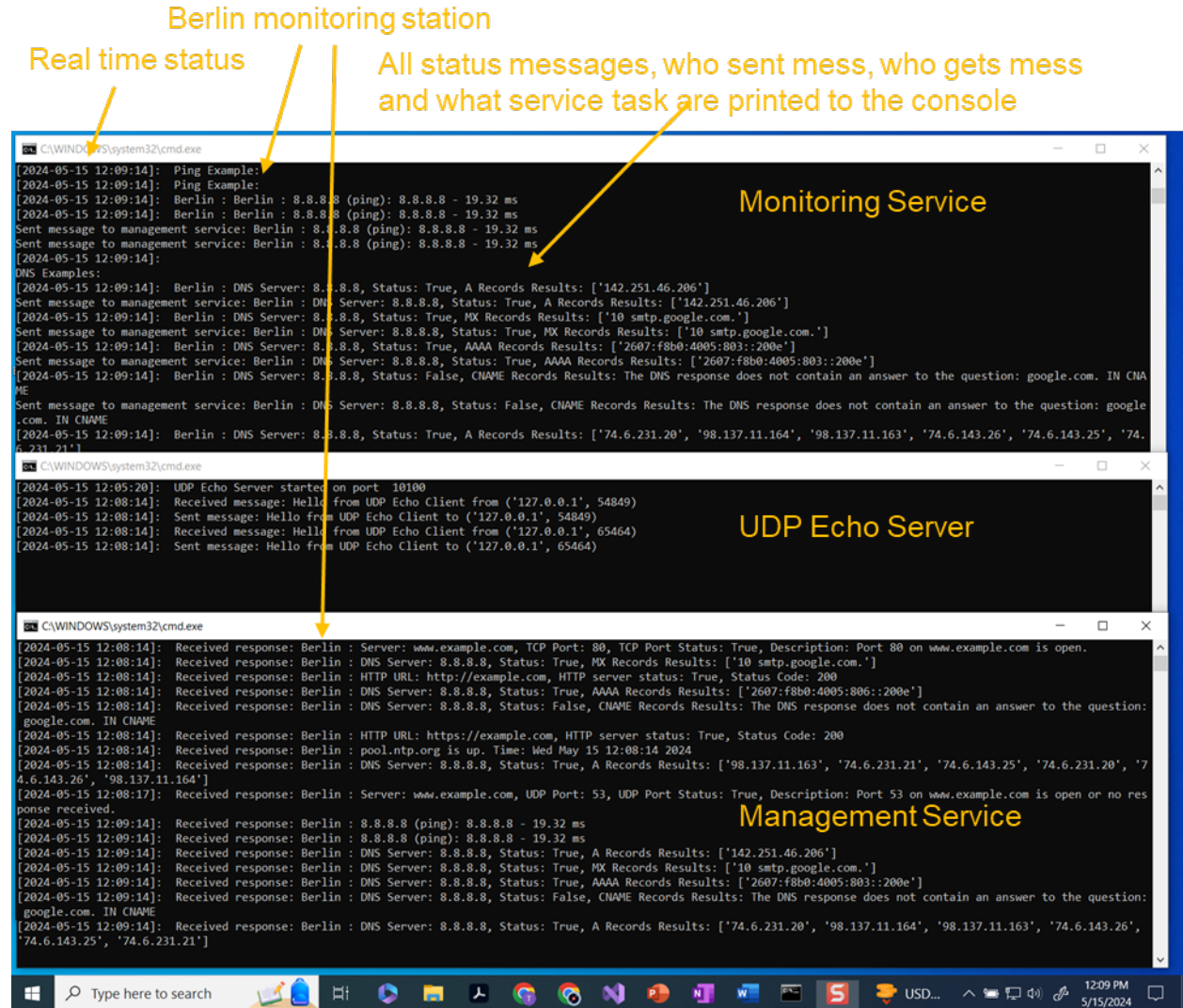
4. Status Display

1. Provide real-time status updates of all connected monitoring services, including their operational status (i.e. validating task information, idle, etc.) and any alerts (i.e. "Online", "Offline", etc.) or anomalies detected (i.e. "Reconnecting", etc.).

1. Print critical messages to the console (or other variant with same goal in GUI).

2. Print status message to console, or provide list upon request, if using command line interface (or other variant with same goal in GUI).

Figure 13



5. Task Distribution and Validation

1. Distribute monitoring tasks to connected monitoring services

Figure 14

From Berlin station monitoring service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:26:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
[2024-05-15 13:27:40]: HTTP/HTTPS Examples:
[2024-05-15 13:27:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:27:40]: DNS Examples:
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
[2024-05-15 13:28:40]: HTTP/HTTPS Examples:
[2024-05-15 13:28:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:28:40]: DNS Examples:
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
[2024-05-15 13:29:40]: TCP Port Example:

```

Monitoring Service

Different monitoring tasks for monitoring service stations.

ping

1 minute period

This is Berlin station monitoring statuses

DNS

1 minute period

Figure 15

```

C:\WINDOWS\system32\cmd.exe
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:33:40]:
TCP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:33:40]:
NTP Example:
[2024-05-15 13:33:40]: Ping Example:
[2024-05-15 13:33:40]: Ping Example:
[2024-05-15 13:33:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
[2024-05-15 13:33:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
[2024-05-15 13:33:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:33:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
[2024-05-15 13:33:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:33:41]:
DNS Examples:
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
[2024-05-15 13:33:46]:
UDP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
[2024-05-15 13:34:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:34:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200

```


Figure 16

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21', '98.137.11.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21', '98.137.11.163']
[2024-05-15 13:37:40]:
TCP Port Example:
[2024-05-15 13:37:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:37:40]:
NTP Examples:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:41]:
DNS Examples:
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:37:42]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164', '98.137.11.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164', '98.137.11.163']
[2024-05-15 13:37:49]:
UDP Port Example:
[2024-05-15 13:37:52]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:42]:

```

ntp
4 minute period

Other monitoring tasks can be
HTTP, HTTPS, ICMP, DNS, NTP, TCP, UDP

1. Retrieve a unique monitoring ID from the Monitoring Service

Berlin Station

Figure 17

Monitoring Service sends its ID, Berlin, with message to Management Service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:37:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 24.71 ms
[2024-05-15 14:37:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['172.217.12.110']
[2024-05-15 14:37:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:37:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 14:37:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: goo
gle.com. IN CNAME
[2024-05-15 14:37:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.
6.143.25', '74.6.231.21']
[2024-05-15 14:38:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 18.35 ms
[2024-05-15 14:38:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 19.32 ms
[2024-05-15 14:38:55]: Received response: Berlin HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:38:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:38:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:38:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
[2024-05-15 14:38:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: goo
gle.com. IN CNAME
[2024-05-15 14:38:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21', '74.6.
231.20', '98.137.11.164']
[2024-05-15 14:39:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 26.40 ms
[2024-05-15 14:39:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 26.40 ms
[2024-05-15 14:39:56]: Received response: Berlin HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:39:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
[2024-05-15 14:39:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:39:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
[2024-05-15 14:39:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: goo
gle.com. IN CNAME
[2024-05-15 14:39:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.13
7.11.164', '74.6.143.25']
[2024-05-15 14:40:55]: Received response: Berlin Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 14:40:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 22.10 ms
[2024-05-15 14:40:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 22.10 ms
[2024-05-15 14:40:56]: Received response: Berlin HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:40:56]: Received response: Berlin pool.ntp.org is up. Time: Wed May 15 14:40:56 2024
[2024-05-15 14:40:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:40:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:40:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:40:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: goo
gle.com. IN CNAME
[2024-05-15 14:40:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21', '74.6.
231.20', '98.137.11.164']
[2024-05-15 14:41:01]: Received response: Berlin Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no respons
e received.
[2024-05-15 14:41:55]: Received response: Berlin Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the messa
ge correctly.
[2024-05-15 14:41:55]: Received response: Berlin Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the messa
ge correctly.
[2024-05-15 14:41:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 16.66 ms
[2024-05-15 14:41:55]: Received response: Berlin 8.8.8.8 (ping): 8.8.8.8 - 21.49 ms
[2024-05-15 14:41:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-15 14:41:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:41:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 14:41:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: goo
gle.com. IN CNAME
[2024-05-15 14:41:56]: Received response: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.
6.143.25', '74.6.231.21']

```

Hong Kong Station

Figure 18

Monitoring Service sends its ID, Hong Kong, with message to Management Service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 18:44:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 17.12 ms
[2024-05-15 18:44:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 22.37 ms
[2024-05-15 18:44:13]: Received response: Hong Kong HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 18:44:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
[2024-05-15 18:44:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 18:44:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 18:44:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question:
google.com. IN CNAME
[2024-05-15 18:44:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.20', '74
.6.231.21', '98.137.11.164']
[2024-05-15 18:45:13]: Received response: Hong Kong Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 18:45:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 25.98 ms
[2024-05-15 18:45:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 21.42 ms
[2024-05-15 18:45:13]: Received response: Hong Kong HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 18:45:13]: Received response: Hong Kong pool.ntp.org is up. Time: Wed May 15 18:45:13 2024
[2024-05-15 18:45:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['172.217.12.110']
[2024-05-15 18:45:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 18:45:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 18:45:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question:
google.com. IN CNAME
[2024-05-15 18:45:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.231.20', '74.6.143.25', '74.6.143.26', '98.1
37.11.164', '98.137.11.163']
[2024-05-15 18:45:19]: Received response: Hong Kong Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no resp
onse received.
[2024-05-15 18:46:13]: Received response: Hong Kong Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the me
ssage correctly.
[2024-05-15 18:46:13]: Received response: Hong Kong Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the me
ssage correctly.
[2024-05-15 18:46:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 17.83 ms
[2024-05-15 18:46:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 17.18 ms
[2024-05-15 18:46:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.46']
[2024-05-15 18:46:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 18:46:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 18:46:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question:
google.com. IN CNAME
[2024-05-15 18:46:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.20', '74
.6.231.21', '98.137.11.164']
[2024-05-15 18:47:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 17.06 ms
[2024-05-15 18:47:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 17.45 ms
[2024-05-15 18:47:13]: Received response: Hong Kong HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 18:47:13]: Received response: Hong Kong HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 18:47:13]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 18:47:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 18:47:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 18:47:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question:
google.com. IN CNAME
[2024-05-15 18:47:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.231.20', '74.6.143.25', '74.6.143.26', '98.1
37.11.164', '98.137.11.163']
[2024-05-15 18:48:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 16.57 ms
[2024-05-15 18:48:13]: Received response: Hong Kong 8.8.8.8 (ping): 8.8.8.8 - 18.38 ms
[2024-05-15 18:48:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['172.217.12.110']
[2024-05-15 18:48:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 18:48:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 18:48:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question:
google.com. IN CNAME
[2024-05-15 18:48:14]: Received response: Hong Kong DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.20', '74

```

2. Receive confirmations from monitoring services for each task, ensuring they have the task configurations.

Figure 19

The Management Service receives confirmation in real time of each task from Berlin Monitoring Service Station
 The Management Station sends out one configuration setup to all Monitoring Stations connected to this Management Station

The figure consists of two screenshots of a Windows command prompt window. The top screenshot shows the Monitoring Service sending messages to the Management Service. The bottom screenshot shows the Management Service receiving responses from the Monitoring Service. A yellow box highlights the 'Received response' lines in the bottom screenshot, and a yellow arrow points from the 'Sent message' lines in the top screenshot to the 'Received response' lines in the bottom screenshot.

Monitoring Service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:42:56]:
HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]:
HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]:
DNS Examples:
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Enter command:

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:41:01]: Received response: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.66 ms
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.40 ms
[2024-05-15 14:41:55]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.21']
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.80 ms
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.77 ms
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
  
```

Management Service

Add beginning of run showing sending and receiving configuration setup Fix***

Monitoring Service Software Requirements Specification (SRS) (Abbreviated):

1. Monitoring Service monitoring actions

1. The Monitoring Service must perform the existing tasks related to running service checks.

Figure 20

From Berlin station monitoring service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:26:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
[2024-05-15 13:27:40]: HTTP/HTTPS Examples:
[2024-05-15 13:27:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:27:40]: DNS Examples:
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
[2024-05-15 13:28:40]: HTTP/HTTPS Examples:
[2024-05-15 13:28:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:28:40]: DNS Examples:
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
[2024-05-15 13:29:40]: TCP Port Example:

```

Monitoring Service

Different monitoring tasks for monitoring service stations.

ping

1 minute period

This is Berlin station monitoring statuses

DNS

1 minute period

Figure 21

Monitoring Service Berlin Station

```

C:\WINDOWS\system32\cmd.exe
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:33:40]:
TCP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:33:40]:
UDP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
[2024-05-15 13:33:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:33:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
[2024-05-15 13:33:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:33:41]:
DNS Examples:
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
[2024-05-15 13:33:46]:
UDP Port Example:
[2024-05-15 13:33:49]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
[2024-05-15 13:34:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:34:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
  
```

ntp and http monitoring tasks

ntp
4 minute period

Figure 22

Monitoring Service Berlin Station

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com
. IN CNAME
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21', '98.137.11
.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21
', '98.137.11.163']
[2024-05-15 13:37:40]:
TCP Port Example:
[2024-05-15 13:37:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:37:40]:
NTP Example:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:41]:
DNS Examples:
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com
. IN CNAME
[2024-05-15 13:37:42]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164', '98.137.11
.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164
', '98.137.11.163']
[2024-05-15 13:37:49]:
UDP Port Example:
[2024-05-15 13:37:52]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response recei
ved.
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:42]:

```

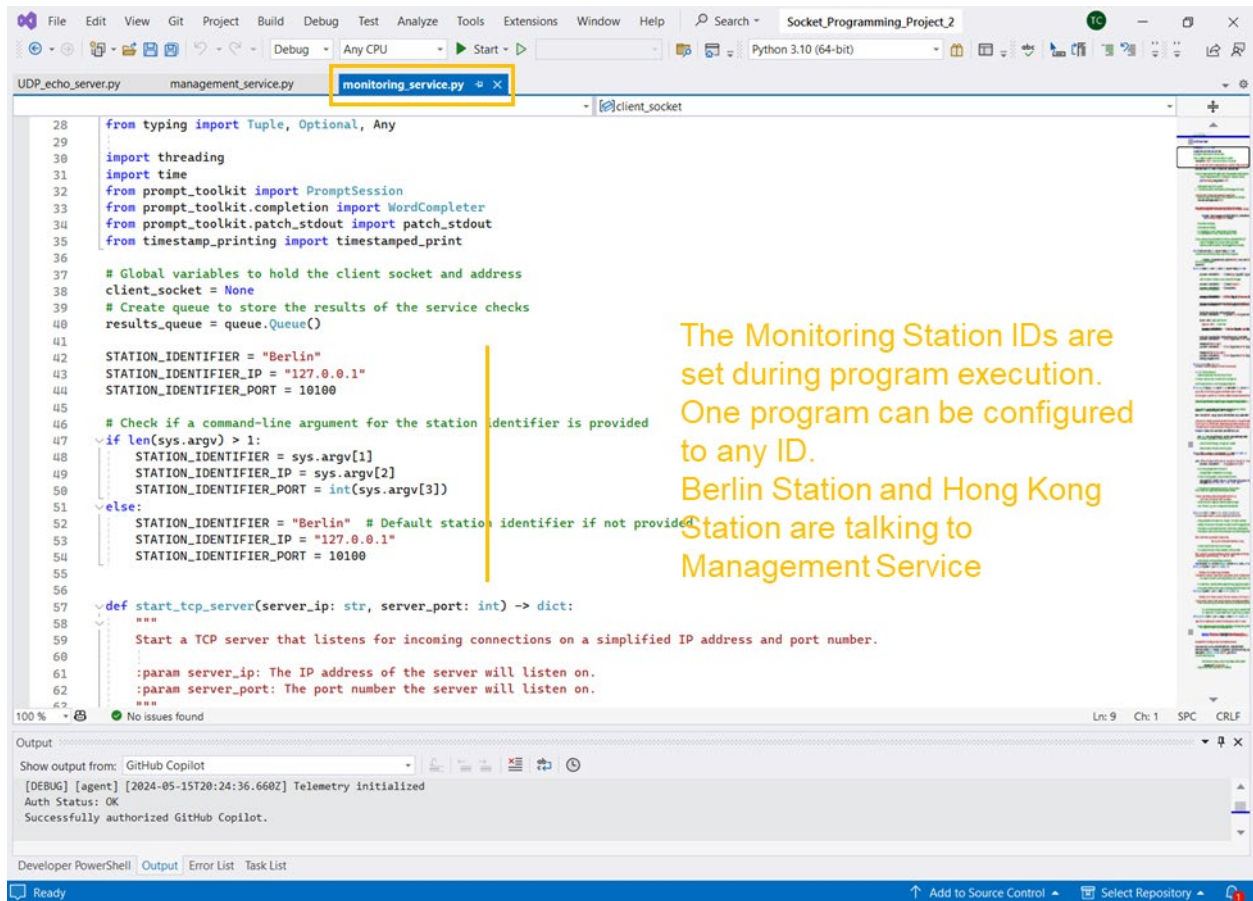
ntp
4 minute period

Other monitoring tasks can be
HTTP, HTTPS, ICMP, DNS, NTP, TCP, UDP

2. Unique Identification

1. Monitoring Service must use a unique identifier for interacting with the Management Service.

Figure 23



3. Task Management

1. Monitoring Service must use unique identifier (R2.1) to perform task management with Management Service.

Berlin Station

Figure 24

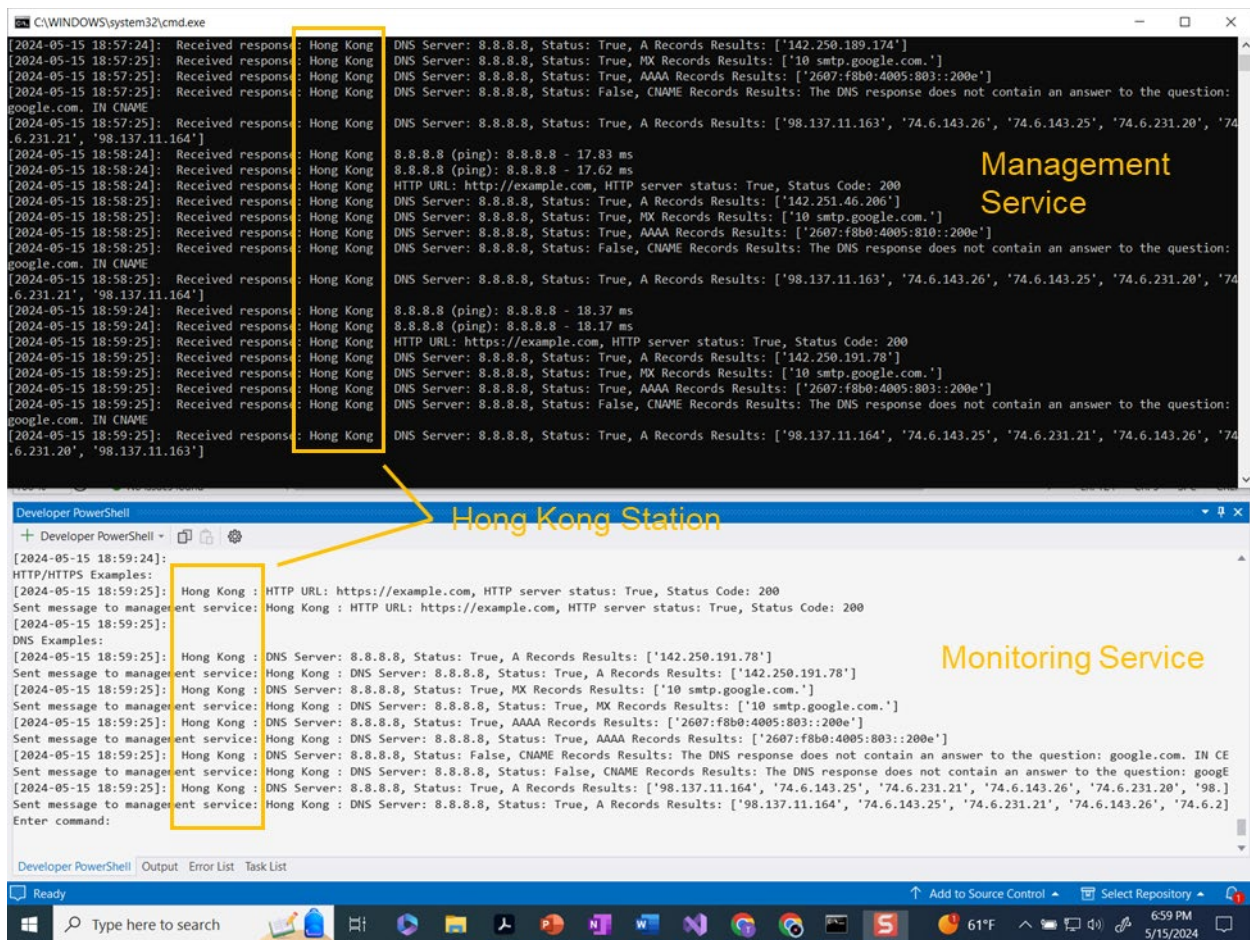
The Management Service knows who is communicating with it by Monitoring Service ID. In this case, Berlin Station is receiving tasks and sending back results.

```
C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: DNS Examples:
[2024-05-15 14:42:56]: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Berlin DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Berlin DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Berlin DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME.
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME.
[2024-05-15 14:42:56]: Berlin DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Enter command:

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:41:01]: Received response: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.66 ms
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.40 ms
[2024-05-15 14:41:55]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.21']
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.80 ms
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.77 ms
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
```

Hong Kong Station

Figure 25



2. Receive all monitoring tasks with the management service.

Figure 26

Berlin Monitoring Station confirms receiving station configuration and sends acknowledgement message back to Management Service

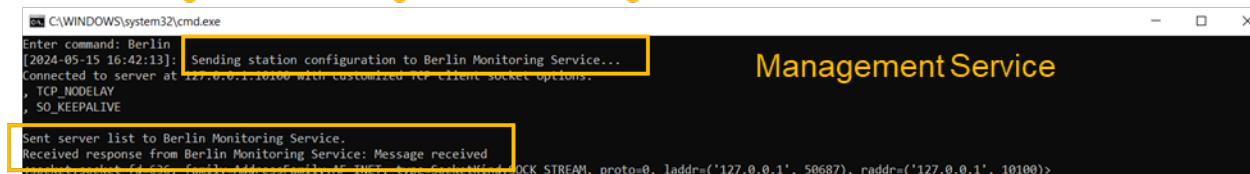


Figure 27

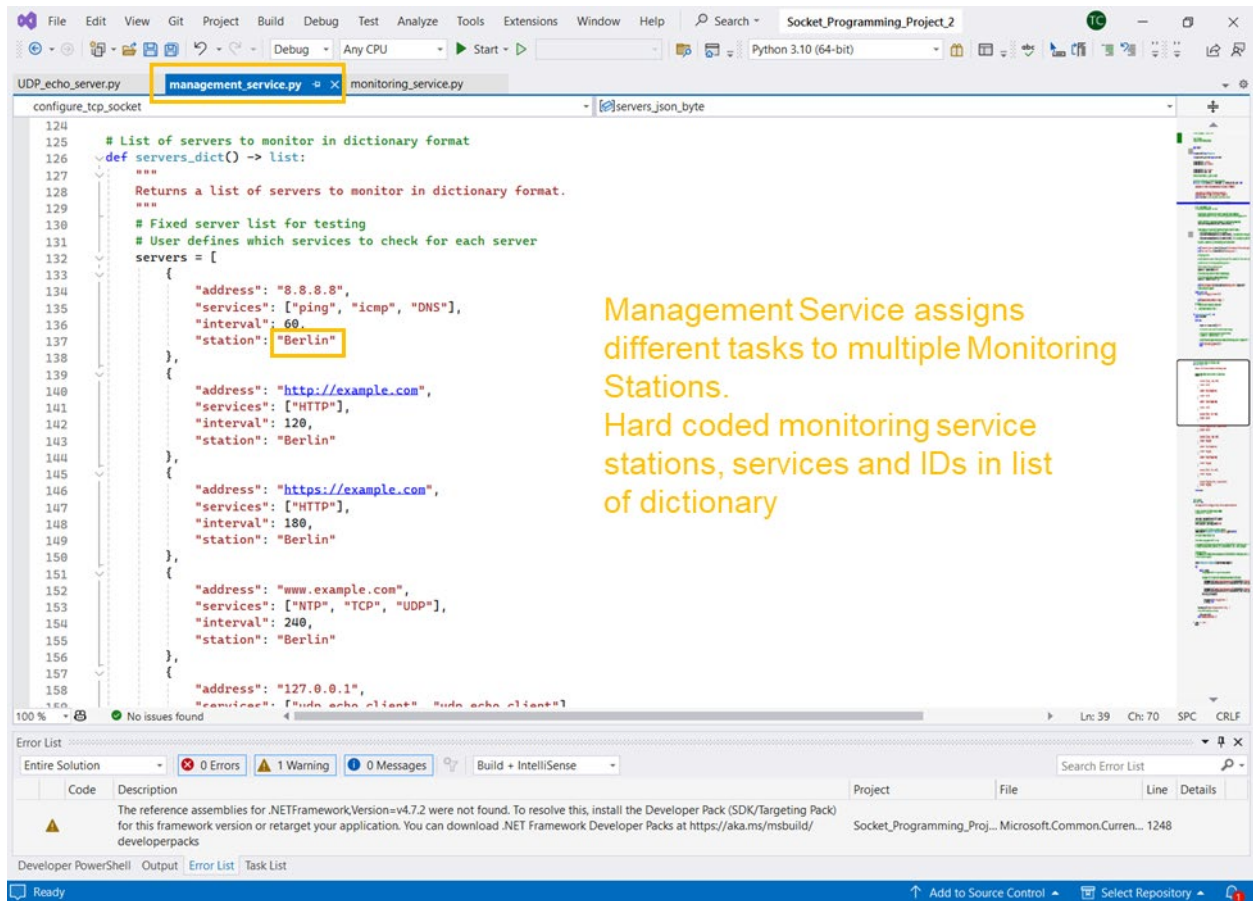
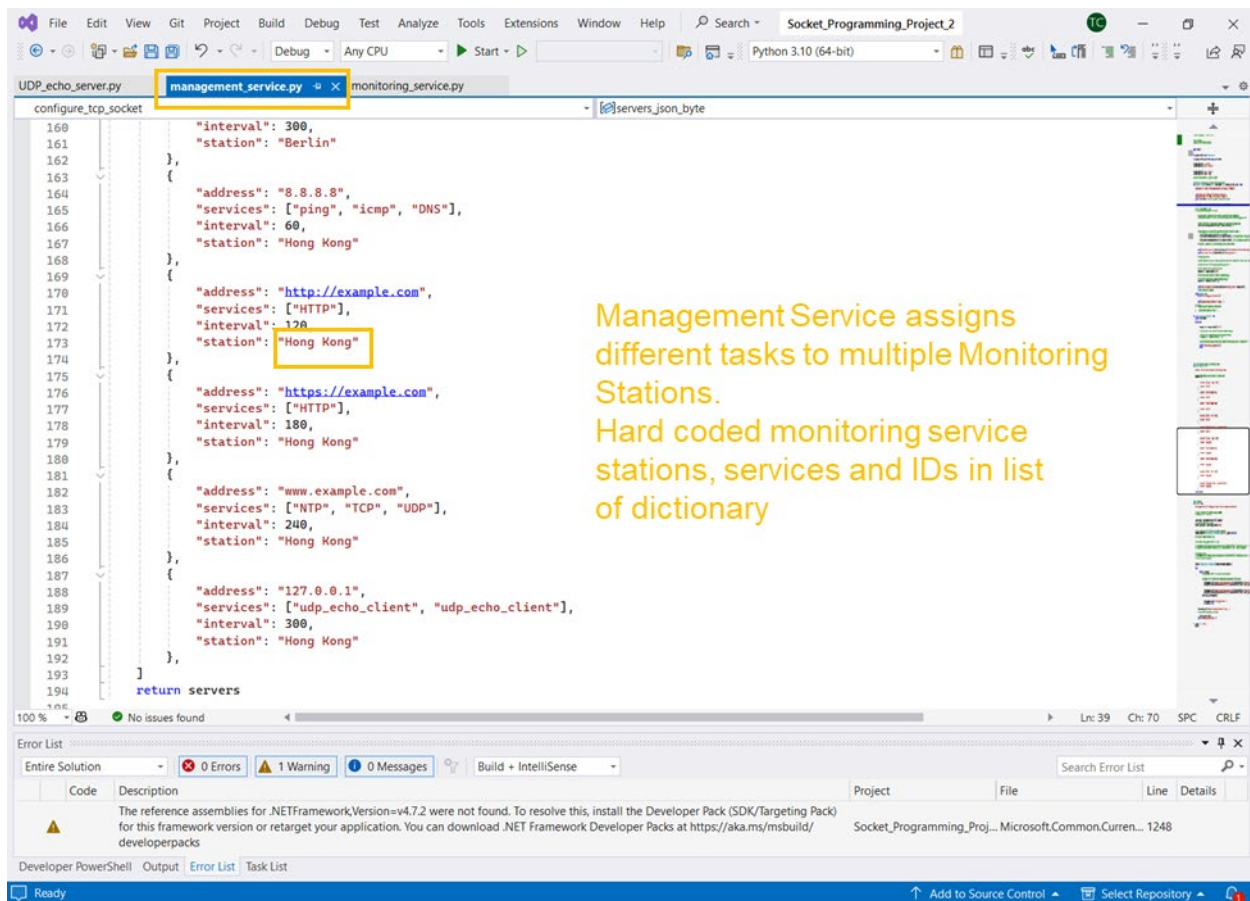


Figure 28



3. Perform assigned monitoring tasks according to the received configurations and frequencies.

Figure 29

From Berlin station monitoring service

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:26:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Ping Example:
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.17 ms
[2024-05-15 13:27:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.89 ms
[2024-05-15 13:27:40]: HTTP/HTTPS Examples:
[2024-05-15 13:27:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:27:40]: DNS Examples:
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.16.206']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:812::200e']
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:27:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Ping Example:
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.55 ms
[2024-05-15 13:28:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.54 ms
[2024-05-15 13:28:40]: HTTP/HTTPS Examples:
[2024-05-15 13:28:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:28:40]: DNS Examples:
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:28:40]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '74.6.143.25', '74.6.143.26', '98.137.11.163', '74.6.231.21']
[2024-05-15 13:29:40]: TCP Port Example:

```

Figure 30

Monitoring Service Berlin Station

```
C:\WINDOWS\system32\cmd.exe
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:811::200e']
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:32:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.163', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.164']
[2024-05-15 13:33:40]:
TCP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:33:40]:
UDP Port Example:
[2024-05-15 13:33:40]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.93 ms
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.93 ms
[2024-05-15 13:33:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:33:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:33:40 2024
[2024-05-15 13:33:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:33:41]:
DNS Examples:
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.46.206']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:810::200e']
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 13:33:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.21', '74.6.143.25', '74.6.231.20', '74.6.143.26', '98.137.11.163']
[2024-05-15 13:33:46]:
UDP Port Example:
[2024-05-15 13:33:49]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 13:34:40]: Ping Example:
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.65 ms
[2024-05-15 13:34:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.29 ms
[2024-05-15 13:34:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:34:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
```

ntp and http monitoring tasks

ntp
4 minute period

Figure 31

Monitoring Service Berlin Station

```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com
. IN CNAME
[2024-05-15 13:36:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21', '98.137.11
.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['98.137.11.164', '74.6.231.20', '74.6.143.25', '74.6.143.26', '74.6.231.21
', '98.137.11.163']
[2024-05-15 13:37:40]:
TCP Port Example:
[2024-05-15 13:37:40]: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
Sent message to management service: Berlin : Server: www.example.com, TCP Port: 80, TCP Port Status: True, Description: Port 80 on www.example.com is open.
[2024-05-15 13:37:40]:
NTP Example:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Ping Example:
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.54 ms
[2024-05-15 13:37:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 22.41 ms
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:40]: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
Sent message to management service: Berlin : pool.ntp.org is up. Time: Wed May 15 13:37:40 2024
[2024-05-15 13:37:40]:
HTTP/HTTPS Examples:
[2024-05-15 13:37:40]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 13:37:41]:
DNS Examples:
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.191.78']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:803::200e']
[2024-05-15 13:37:41]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com
. IN CNAME
[2024-05-15 13:37:42]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164', '98.137.11
.163']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.21', '74.6.143.25', '74.6.143.26', '74.6.231.20', '98.137.11.164
', '98.137.11.163']
[2024-05-15 13:37:49]:
UDP Port Example:
[2024-05-15 13:37:52]: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
Sent message to management service: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response recei
ved.
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Ping Example:
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:40]: Berlin : Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
Sent message to management service: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 20.11 ms
[2024-05-15 13:38:42]:

```

ntp
4 minute period

Other monitoring tasks can be
HTTP, HTTPS, ICMP, DNS, NTP, TCP, UDP

4. Result Reporting

1. Monitoring Service must use unique identifier (R2.1) to perform result reporting with Management Service.

Combine 4.1 and 4.2.

2. Send the results of service tests back to the management service immediately for real-time reporting.

Figure 32

The Management Service receives confirmation in real time of each task from Berlin Monitoring Service Station
 The Management Station sends out one configuration setup to all Monitoring Stations connected to this Management Station

Monitoring Service

Monitoring results here

Results are sent back to Management Service

and Management results here

Management Service

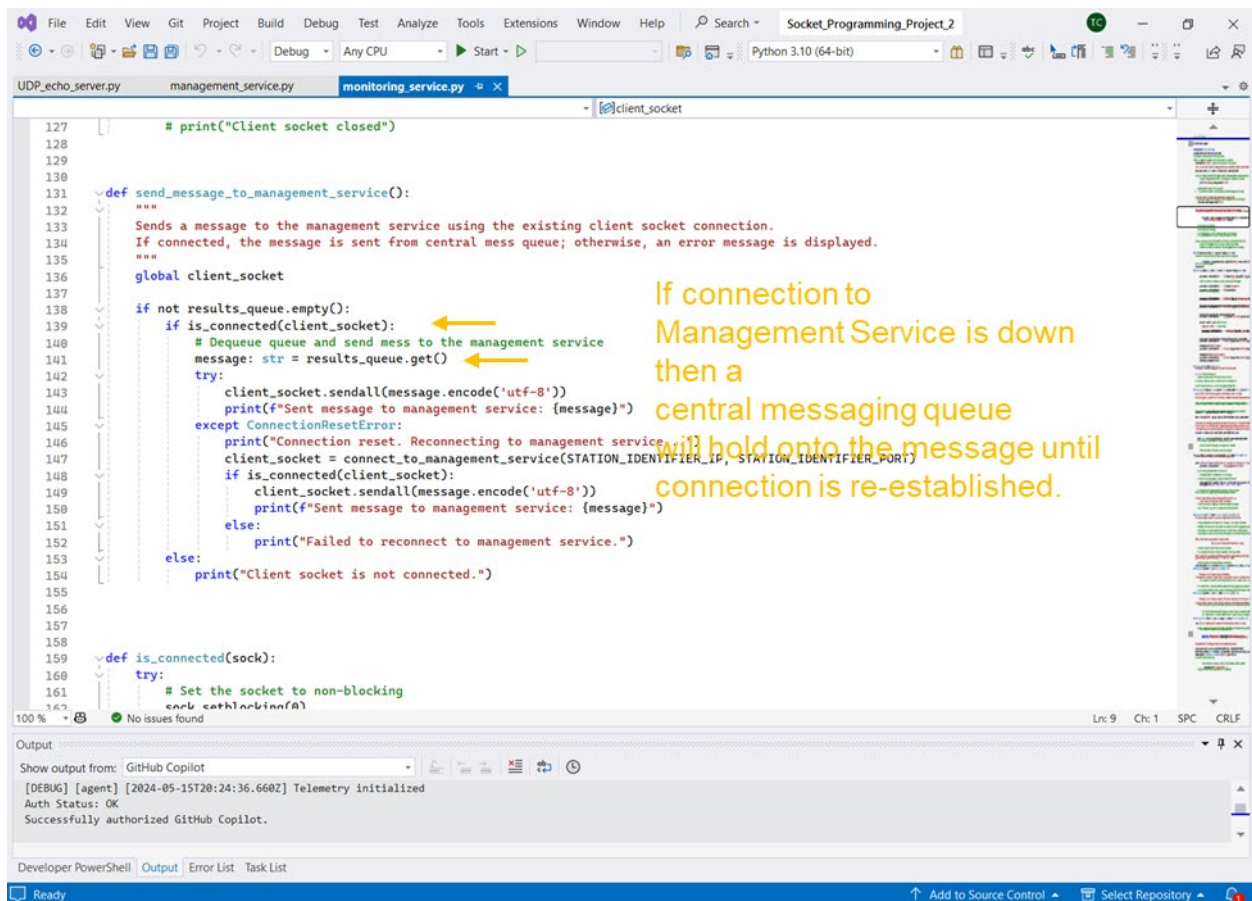
```

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: DNS Examples:
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Enter command:

C:\WINDOWS\system32\cmd.exe
[2024-05-15 14:41:01]: Received response: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 of www.example.com is open or no response received.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.66 ms
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.40 ms
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.21']
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.80 ms
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.77 ms
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
  
```

1. If connection is down, hold onto results until next connection from Management Service.

Figure 33



The status results of all the Monitor Service tasks go to a central message queue. Before the messages are sent across the socket to the Managing Service, the socket connection is checked for a valid connection. If no valid connection is present then the queue waits until there is a valid connection. The central messaging queue is a FIFO.

3. Maintain timestamps for all monitoring results (transferred to Management Service as part of the result for accurate reporting).

Figure 34

The Management Service will use the same time stamp and status message as Monitoring Service

The figure consists of two screenshots of a Windows command prompt window, showing logs for two different services: Monitoring Service and Management Service. Both services are running on a system with IP 127.0.0.1 and port 10100. The logs show that both services are performing the same set of tests: HTTP, DNS, and ping. The logs for both services are identical, indicating that the Management Service is using the same time stamp and status message as the Monitoring Service.

Monitoring Service Log:

```
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: HTTP/HTTPS Examples:
[2024-05-15 14:42:56]: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
Sent message to management service: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: DNS Examples:
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Sent message to management service: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
Enter command:
```

Management Service Log:

```
[2024-05-15 14:41:01]: Received response: Berlin : Server: www.example.com, UDP Port: 53, UDP Port Status: True, Description: Port 53 on www.example.com is open or no response received.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : Server: 127.0.0.1, UDP Port: 10100, UDP Port Status: True, Description: UDP echo server at 127.0.0.1:10100 echoed the message correctly.
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 16.66 ms
[2024-05-15 14:41:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 21.49 ms
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.250.188.14']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:41:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.231.20', '98.137.11.164', '98.137.11.163', '74.6.143.26', '74.6.143.25', '74.6.231.21']
[2024-05-15 14:41:56]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 19.80 ms
[2024-05-15 14:42:55]: Received response: Berlin : 8.8.8.8 (ping): 8.8.8.8 - 18.77 ms
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: http://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : HTTP URL: https://example.com, HTTP server status: True, Status Code: 200
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['142.251.32.46']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, MX Records Results: ['10 smtp.google.com.']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, AAAA Records Results: ['2607:f8b0:4005:806::200e']
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: False, CNAME Records Results: The DNS response does not contain an answer to the question: google.com. IN CNAME
[2024-05-15 14:42:56]: Received response: Berlin : DNS Server: 8.8.8.8, Status: True, A Records Results: ['74.6.143.26', '74.6.231.20', '74.6.231.21', '98.137.11.163', '98.137.11.164', '74.6.143.25']
```

5. Resilience and Recovery

1. Monitoring Service must be able to allow for the Management Service to reconnect without needing to restart the Monitoring Service.

Figure 35

